Hyperbola

Mind map

Recognition

- $\Delta \neq 0,\ h^2 > ab$
- $|PS_1 - PS_2| = 2a$

Standard form

- $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
- $b^2 = a^2(e^2 - 1)$
- $\ell_{LR} = \frac{2b^2}{a}$

Focal distances

- $|ex_1 \mp a|$
- Difference $= 2a$

Auxiliary circle

- $x^2 + y^2 = a^2$
- $(a\sec\theta, b\tan\theta)$

Shifted / other forms

- Complete squares
- $\frac{p_2^2}{a^2} - \frac{p_1^2}{b^2} = 1$

Conjugate

- Constant $\to -1$
- $\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$

Rectangular

- $a = b,\ e = \sqrt{2}$
- LR = TA = CA

Point and chord

- $S_1 > 0$ inside
- Chord $P(\alpha)Q(\beta)$

Line and hyperbola

- $c^2 = a^2m^2 - b^2$
- Need $|m| > \frac{b}{a}$

Director circle

- $x^2 + y^2 = a^2 - b^2$
- Real only if $a > b$

Tangent

- $T = 0$
- $\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$
- $\left(-\frac{a^2m}{c}, -\frac{b^2}{c}\right)$

Normal

- $\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$
- Up to 4 from a point

Chord family

- $T = S_1$
- $SS_1 = T^2$
- Pole and polar

Highlights

- $p_1p_2 = b^2$
- Feet on aux. circle
- Confocal $\perp$

Asymptotes

- $H + C = 2A$
- $e = \sec\frac{\theta}{2}$
- Area $CQR = ab$

$xy = c^2$

- $\left(ct, \frac{c}{t}\right)$
- $t^3t_1 = -1$
- $\frac{x}{h} + \frac{y}{k} = 2$